

PADHAISHURU

GATE Preparation Platform · padhaishuru.com

TREND-BASED MOCK PAPER

GATE Metallurgical Engineering

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Metallurgical Engineering
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

Unofficial Practice Material — Not affiliated with or endorsed by GATE/IIT.

Prepared by PadhaiShuru (padhaishuru.com) for practice purposes only.

Section A: General Aptitude

Q1. [1-MCQ] [1 mark]

10% of 500:

- (A) 5
- (B) 50
- (C) 500
- (D) 0.5

Section B: Metallurgical Engineering

Q2. [1-MCQ] [1 mark]

Rank of 3x3 zero matrix:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q3. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q4. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr

(D) $2\pi r^2$

Q5. [1-MCQ] [1 mark]

Which of the following is a probability generating function for Poisson(λ)?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q6. [1-MCQ] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q7. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1,1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q8. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q9. [1-NAT] [1 mark]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q10. [1-MCQ] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q11. [1-MCQ] [1 mark]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q12. [1-MCQ] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

- (A) $y = C \cdot x$
- (B) $y = C/x$
- (C) $y = C \cdot x^2$
- (D) $x^2 + y^2 = C$

Q13. [1-MCQ] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq$:

- (A) $\|u\| \cdot \|v\|$
- (B) $\|u\| + \|v\|$
- (C) $\|u\| / \|v\|$
- (D) $\sqrt{\|u\| \cdot \|v\|}$

Q14. [2-MCQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q15. [2-NAT] [2 marks]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T$ equals:

(Numerical Answer Type — Enter the numerical value)

Q16. [2-MCQ] [2 marks]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e
- (B) e^2
- (C) 1
- (D) infinity

Q17. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi, \pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q18. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q19. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant

(D) Non-singular

Q20. [2-MCQ] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q21. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q22. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q23. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q24. [2-MCQ] [2 marks]

The Laplace transform of $t \cdot e^{(-2t)}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q25. [2-MSQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2
- (B) 0.5
- (C) 0.6
- (D) 0.4

Q26. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q27. [2-NAT] [2 marks]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

(Numerical Answer Type — Enter the numerical value)

Q28. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

- (A) 1
 - (B) 2
 - (C) 3
 - (D) 0
-

Q29. [2-MSQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
 - (B) 2
 - (C) 16
 - (D) 1
-

Q30. [2-MSQ] [2 marks]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
 - (B) 0.32
 - (C) 0.95
 - (D) 0.16
-

Q31. [1-MCQ] [1 mark]

Reducing agent in blast furnace:

- (A) Coke
 - (B) Limestone
 - (C) Hot air
 - (D) Slag
-

Q32. [1-NAT] [1 mark]

Blast furnace main product:

(Numerical Answer Type — Enter the numerical value)

Q33. [1-MCQ] [1 mark]

Halls-Heroult process produces:

- (A) Iron
 - (B) Aluminum
 - (C) Copper
 - (D) Zinc
-

Q34. [2-NAT] [2 marks]

Blast furnace main product:

(Numerical Answer Type — Enter the numerical value)

Q35. [2-MCQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron

- (B) Aluminum
- (C) Copper
- (D) Zinc

Q36. [2-MSQ] [2 marks]

Blast furnace main product:

- (A) Steel
- (B) Pig iron
- (C) Copper
- (D) Aluminum

Q37. [2-MCQ] [2 marks]

Blast furnace main product:

- (A) Steel
- (B) Pig iron
- (C) Copper
- (D) Aluminum

Q38. [2-MCQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

Q39. [2-NAT] [2 marks]

Blast furnace main product:

(Numerical Answer Type — Enter the numerical value)

Q40. [2-MSQ] [2 marks]

Halls-Heroult process produces:

- (A) Iron
- (B) Aluminum
- (C) Copper
- (D) Zinc

Q41. [1-NAT] [1 mark]

BCC coordination number:

(Numerical Answer Type — Enter the numerical value)

Q42. [1-MCQ] [1 mark]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q43. [1-NAT] [1 mark]

Eutectic reaction:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q45. [2-MCQ] [2 marks]

Phase diagram lever rule finds:

- (A) Temperature
- (B) Phase fraction
- (C) Composition
- (D) Density

Q46. [2-MCQ] [2 marks]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q47. [2-NAT] [2 marks]

BCC structure coordination number:

(Numerical Answer Type — Enter the numerical value)

Q48. [2-MCQ] [2 marks]

Eutectic reaction:

- (A) Liquid $\rightarrow$ α + β
- (B) $\alpha \rightarrow \beta$ + γ
- (C) Liquid $\rightarrow \alpha$
- (D) α + $\beta \rightarrow \gamma$

Q49. [2-MCQ] [2 marks]

BCC structure coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q50. [2-NAT] [2 marks]

Phase diagram lever rule finds:

(Numerical Answer Type — Enter the numerical value)

Q51. [1-MCQ] [1 mark]

Brass is:

- (A) Cu-Zn
- (B) Cu-Sn
- (C) Cu-Ni
- (D) Cu-Al

Q52. [1-MCQ] [1 mark]

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

Q53. [1-MCQ] [1 mark]

BCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q54. [1-MCQ] [1 mark]

FCC coordination number:

- (A) 6
- (B) 8
- (C) 12
- (D) 4

Q55. [2-MCQ] [2 marks]

Brass is:

- (A) Cu-Zn
- (B) Cu-Sn
- (C) Cu-Ni
- (D) Cu-Al

Q56. [2-NAT] [2 marks]

BCC coordination number:

(Numerical Answer Type — Enter the numerical value)

Q57. [2-MCQ] [2 marks]

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

Q58. [2-NAT] [2 marks]

FCC coordination number:

(Numerical Answer Type — Enter the numerical value)

Q59. [2-MCQ] [2 marks]

Steel hardening process:

- (A) Annealing
- (B) Quenching
- (C) Normalizing
- (D) Tempering

Q60. [1-MCQ] [1 mark]

Edge dislocation moves by:

- (A) Climb
- (B) Glide
- (C) Cross slip
- (D) None

Q61. [1-MCQ] [1 mark]

Eutectic point has:

- (A) One phase
- (B) Two phases
- (C) Three phases simultaneously
- (D) Four phases

Q62. [1-MCQ] [1 mark]

Martensite is formed by:

- (A) Slow cooling
- (B) Quenching
- (C) Annealing
- (D) Normalizing

Q63. [1-MCQ] [1 mark]

Cathodic reaction in rusting:

- (A) $\text{Fe} \rightarrow \text{Fe}^{2+} + 2\text{e}^-$
- (B) $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$
- (C) Both
- (D) None

Q64. [1-MCQ] [1 mark]

Shielded metal arc uses:

- (A) Gas shield
- (B) Flux coated electrode
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Duralumin is Al +:

- (A) Cu
 - (B) Mg
 - (C) Si
 - (D) Mn
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1MCQ	B	1
2	1MCQ	A	1
3	1NAT	C	1
4	1MCQ	B	1
5	1MCQ	A	1
6	1MCQ	C	1
7	1MCQ	B	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	A	1
11	1MCQ	B	1
12	1MCQ	A	1
13	1NAT	A	1

14	1MCQ	B	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1MCQ	A	1
18	1MCQ	A	1
19	1NAT	B	1
20	1MCQ	A	1
21	1MCQ	B	1
22	1MCQ	B	1
23	1MCQ	B	1
24	1MCQ	C	1
25	1MCQ	C	1
26	1NAT	C	1
27	1MCQ	A	1
28	1MCQ	A	1

29	1MCQ	B	1
30	1MCQ	A	1
31	2MCQ	A	2
32	2NAT	A	2
33	2MCQ	B	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2MCQ	C	2
37	2MCQ	C	2
38	2MCQ	B	2
39	2MCQ	B	2
40	2MCQ	A	2
41	2MCQ	A	2
42	2MSQ	B	2
43	2MCQ	C	2

44	2NAT	A	2
45	2NAT	B	2
46	2MCQ	A	2
47	2MCQ	B	2
48	2MCQ	B	2
49	2MSQ	A	2
50	2MCQ	B	2
51	2NAT	B	2
52	2MCQ	A	2
53	2MSQ	B	2
54	2NAT	B	2
55	2MSQ	A	2
56	2MCQ	B	2
57	2MCQ	B	2
58	2MCQ	A	2

59	2MCQ	B	2
60	2NAT	B	2
61	2NAT	C	2
62	2MCQ	B	2
63	2MSQ	B	2
64	2NAT	B	2
65	2MCQ	B	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: moderate

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: moderate

Q3.

Answer: C

Solution:

Correct Answer: C

Subject: Physical Metallurgy | Topic: Crystal Structure | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Mechanical Metallurgy | Topic: Dislocations | Difficulty: moderate

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Extractive Metallurgy | Topic: Iron Making | Difficulty: easy

Q6.

Answer: C

Solution:

Correct Answer: C

Subject: Phase Transformations | Topic: Phase Diagrams | Difficulty: easy

Q7.

Answer: B

Solution:

Correct Answer: B

Subject: Heat Treatment | Topic: Steel | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Corrosion | Topic: Electrochemical | Difficulty: difficult

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Welding & Joining | Topic: Arc Welding | Difficulty: easy

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Non-Ferrous Metals & Alloys | Topic: Aluminum | Difficulty: moderate

Q11.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q12.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q13.

Answer: A

Solution:

Eutectic: $L \rightarrow \alpha + \beta$ at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q14.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q15.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q16.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q17.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q19.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone $\rightarrow$ pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q20.

Answer: A

Solution:

Row 2 = $2 \times$ Row 1, Row 3 = $3 \times$ Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q21.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q22.

Answer: B

Solution:

BCC: 8 nearest neighbors. Coordination number = 8.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q23.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q24.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q25.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q26.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q28.

Answer: A

Solution:

Poisson: mean = λ = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q29.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C \cdot x$. Wait: $y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C \cdot x$. That's option A. Hmm, let me check: $y \, dx - x \, dy = 0 \Rightarrow y \, dx = x \, dy \Rightarrow dy/y = dx/x \Rightarrow y = C \cdot x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q30.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| \cdot \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1) \approx 0.68$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50} [1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q33.

Answer: B

Solution:

$$(1+1/n)^{2n} = [(1+1/n)^n]^2 \rightarrow e^2 \text{ as } n \rightarrow \infty.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: A

Solution:

$$f(i) = i^2 + 3i + 2 = -1 + 3i. \operatorname{Re}(f(i)) = -1.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q37.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: B

Solution:

$$\operatorname{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0).$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q39.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q40.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q41.

Answer: A

Solution:

$$L\{t * f(t)\} = -d/ds(F(s)). F(s) = 1/(s+2). -d/ds[1/(s+2)] = 1/(s+2)^2.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q42.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q43.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q45.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone -> pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q46.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q48.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q49.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q50.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q51.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q52.

Answer: A

Solution:

Brass: copper-zinc alloy. Bronze: copper-tin.

Subject: Material Science | Topic: Material Science | Difficulty: difficult

Q53.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone -> pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q54.

Answer: B

Solution:

BCC: 8 nearest neighbors. Coordination number = 8.

Subject: Material Science | Topic: Material Science | Difficulty: moderate

Q55.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone $\rightarrow$ pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q57.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q58.

Answer: A

Solution:

Eutectic: L $\rightarrow$ alpha + beta at eutectic temperature. One liquid to two solids.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q59.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q60.

Answer: B

Solution:

Blast furnace: iron ore + coke + limestone $\rightarrow$ pig iron (high carbon iron).

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: difficult

Q61.

Answer: C

Solution:

FCC: 12 nearest neighbors. Coordination number = 12.

Subject: Material Science | Topic: Material Science | Difficulty: easy

Q62.

Answer: B

Solution:

BCC: 8 nearest neighbors at cube corners.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: moderate

Q63.

Answer: B

Solution:

Hall-Heroult: electrolytic reduction of alumina (Al_2O_3) to aluminum.

Subject: Extractive Metallurgy | Topic: Extractive Metallurgy | Difficulty: moderate

Q64.

Answer: B

Solution:

Lever rule: fraction of phase = opposite arm / total tie-line length.

Subject: Physical Metallurgy | Topic: Physical Metallurgy | Difficulty: easy

Q65.

Answer: B

Solution:

Quenching: rapid cooling from austenitizing temp. Forms hard martensite.

Subject: Material Science | Topic: Material Science | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 29 questions, 46 marks
Extractive Metallurgy: 10 questions, 17 marks
Physical Metallurgy: 10 questions, 16 marks
Material Science: 9 questions, 14 marks
General Aptitude: 1 questions, 1 marks
Mechanical Metallurgy: 1 questions, 1 marks
Phase Transformations: 1 questions, 1 marks
Heat Treatment: 1 questions, 1 marks
Corrosion: 1 questions, 1 marks
Welding & Joining: 1 questions, 1 marks
Non-Ferrous Metals & Alloys: 1 questions, 1 marks

Difficulty Distribution:

Easy: 15 | Moderate: 40 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.